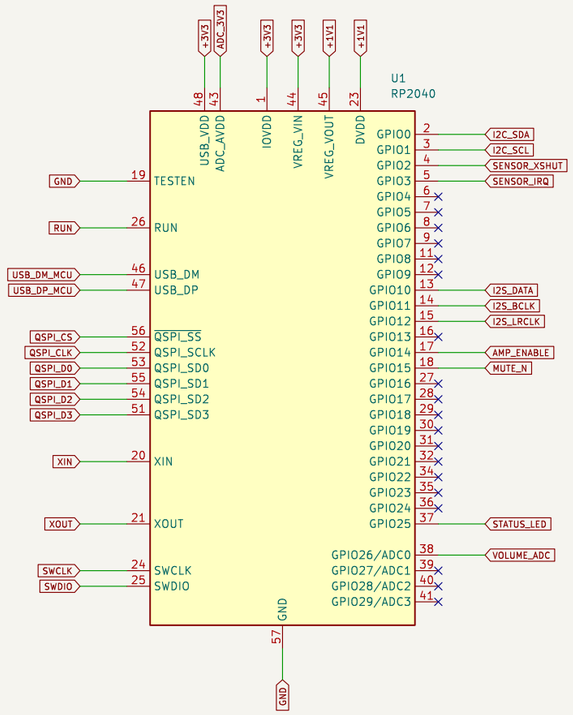
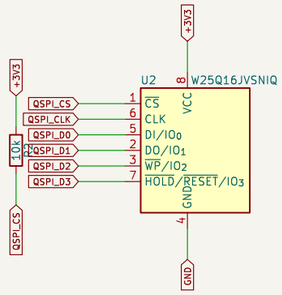
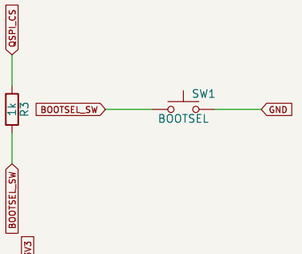
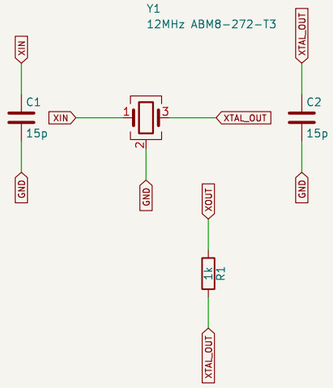
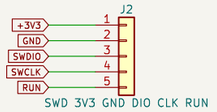






Reference: Raspberry Pi Hardware design with RP2040, sections 2.1–2.5.

One 100 nF capacitor per MCU supply pin: +1V1 from on-chip regulator.

12 MHz ABM8-272-T3; 1 kohm drive resistor and 15 pF load capacitors.

Prototype only: assembly, factory quote and physical validation pending.

Power

File: power.kicad_sch

Instrument

File: instrument.kicad_sch

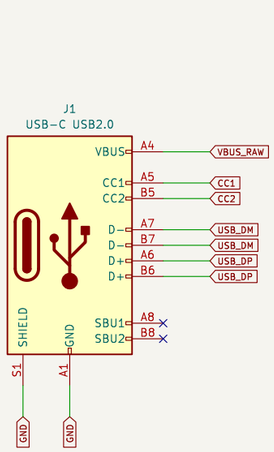
**AirNote side project**

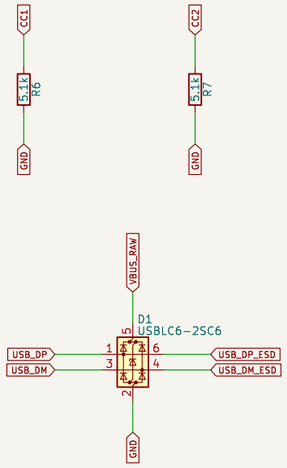
Sheet: /

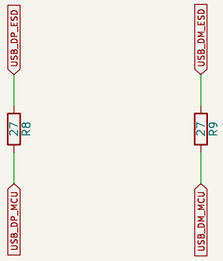
File: airnote.kicad_sch

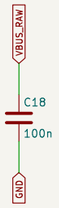
***Title: AirNote – RP2040 controller***

| Size: A3 | Date: 2026-10-10 | **Rev: 0.1 DRAFT** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/3 |



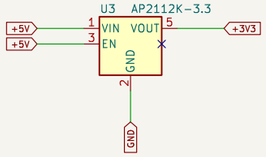











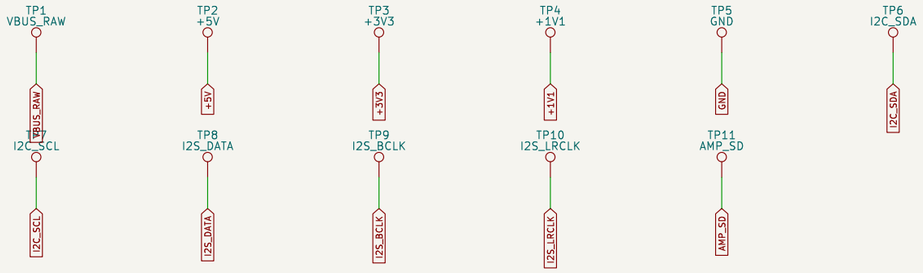











USB wall adapter: regulated 5 V, >=1 A. ONE SUPPLY PATH ONLY.

Computer programming: set hardware switch to MUTE before connection.

USB-C: separate 5.1k Rd on CC1 and CC2; USB 2.0 data only, no PD.

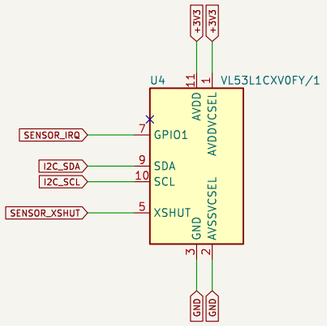
3V3 LDO thermal/current budget and USB rail drop require bench verification.

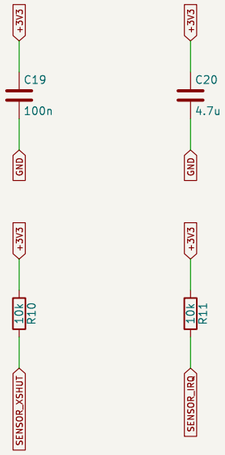
**AirNote side project**

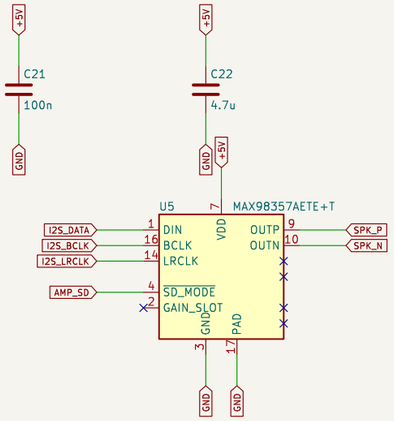
Sheet: /Power/
File: power.kicad_sch

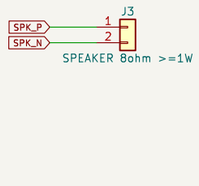
***Title: AirNote — USB and power***

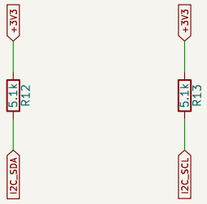
| Size: A3 | Date: 2026-10-10 | **Rev: 0.1 DRAFT** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 2/3 |

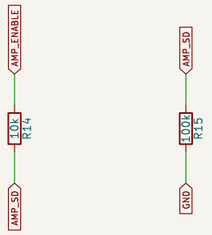


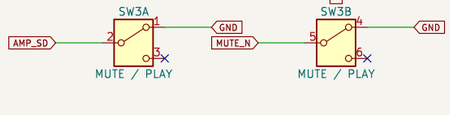



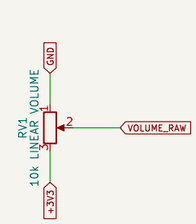



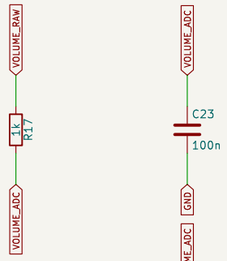



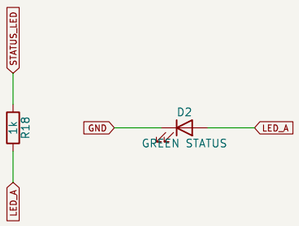










Sensor AVDD and pull-ups = 3.3 V. Driver must explicitly select 2V8 I/O mode.

XSHUT: drive low or release high-Z. Never enable RP2040 internal pull-up.

MAX98357A: GAIN floating = 9 dB; SD high selects LEFT. Duplicate audio on L/R.

Hardware MUTE grounds AMP_SD through switch and drives MUTE_N low.

Speaker is bridge-tied: neither speaker wire may connect to ground.

Optical factory process: dry reflow, no-wash, MSL3, remove liner, offset calibrate.

**AirNote side project**

Sheet: /Instrument/
File: instrument.kicad_sch

***Title: AirNote — sensor, audio and controls***

| Size: A3 | Date: 2026-10-10 | **Rev: 0.1 DRAFT** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 3/3 |